In-Class Exercise 2 · Solutions

Jigsaw latch • 95% lower CI (instructor key)

Summaries.

$$n = 10, \quad \bar{x} = 52.5, \quad s = \sqrt{\frac{1}{9}\sum (x_i - \bar{x})^2} \approx 3.028.$$

Critical value. One-sided 95% lower bound $\Rightarrow \alpha = 0.05$, df $= n - 1 = 9$:

$$t_{0.05}(9) = 1.833.$$

Lower bound.

$$L = \bar{x} - t_{0.05}(9)\,\frac{s}{\sqrt{n}} = 52.5 - 1.833 \times \frac{3.028}{\sqrt{10}} \approx 52.5 - 1.755 \approx 50.745.$$

Interval:

$$[\,50.745,\ \infty\,).$$

Claim $\mu \geq 50$. The entire lower CI lies above 50 ($L > 50$), so at the 95% level the data support the advertising claim. (Equivalently: a level-$\alpha = 0.05$ test of $H_0 : \mu \leq 50$ vs $H_1 : \mu > 50$ rejects H_0.)

Common trap. Using $t_{0.025}(9) = 2.262$ (two-sided) gives a lower L and the wrong critical value for a one-sided bound.

Quick check

$L \approx$ **50.75** • interval $[50.75, \infty)$ • claim supported • Jigsaw lets you go.